CAMBRIDGE O LEVEL COMPUTER SCIENCE (2210 / 0478)

Paper 1: Computer Systems | Topic-Wise Question Bank (2024 – 2026 Series)

TOPIC 04: SOFTWARE

Exam Sources: Cambridge 2210 / 0478 (May/June 2023, Oct/Nov 2023, May/June 2024)

Question 1 [2210/12/M/J/24 Q4 (2024 Series)] [Total: 6 marks]

(a) Distinguish between system software and application software. Give two examples of each. **[3 marks]**

(b) State three key functions of an Operating System (OS). **[3 marks]**

✓ Marking Scheme / Model Answer:

(a) System software manages computer hardware and system operations (e.g. Windows OS, Device drivers, Anti-virus utility) [1.5 marks]. Application software allows users to perform end-user tasks (e.g. Word processor, Web browser, Photoshop) [1.5 marks].
(b) OS Functions: Memory management, Multitasking/Process scheduling, Device management via drivers, File system management, User authentication/Security [3 marks].

Question 2 [2210/12/M/J/23 Q3 (2023 Series)] [Total: 6 marks]

(a) Compare high-level programming languages and low-level programming languages with reference to:

(i) Machine independence / portability
(ii) Ease of reading, writing, and debugging
(iii) Execution efficiency **[3 marks]**

(b) Explain the role of a compiler, an interpreter, and an assembler. **[3 marks]**

✓ Marking Scheme / Model Answer:

(a) High-level languages: Portable across different CPU architectures; human-friendly syntax; easier to debug [1.5 marks]. Low-level languages: Specific to CPU architecture; direct hardware memory control; faster execution and compact machine code [1.5 marks].
(b) Compiler: Translates whole high-level program into machine code at once, creating standalone executable file [1 mark].
Interpreter: Translates and executes high-level code line-by-line in real-time [1 mark].
Assembler: Translates Assembly language (mnemonics) into binary machine code [1 mark].
